NC SUD Application Study: Completed Results

Treatment assignment designs for a spatial cluster randomized education pilot on observed North Carolina SUD incidence, 2018–2021

Andrew Walther

October 12, 2026

Table of contents

Executive summary 2

1 Background: what the grid simulation showed 3

2 Data and preparation 3

3 Clusters and spatial weights 4

4 Adapting the designs to the NC map 4

5 Methods 5

6 Results 7

7 Comparison with the grid findings 9

8 Implications and limitations 9

9 Additional results 10

Key decisions made 12

Executive summary

Question. When an education intervention for healthcare professionals is randomized across North Carolina's (NC's) 58 community-college service areas, which treatment assignment design estimates the intervention effect most accurately, given spatial spillover and the option of using historical sudden unexpected death (SUD) incidence to direct treatment?

- **Data.** Restricted county-level sudden unexpected death (SUD) death-certificate counts (A. Habib) and Surveillance, Epidemiology, and End Results (SEER) program populations aged 18–64 for 2018–2021: 23,523 deaths over 25,594,321 person-years, aggregated to 58 service-area clusters. Statewide rates rose from 83.9 (2018) to 100.5 (2021) per 100,000; pooled cluster rates span 45.3–194.9. Each year is a fixed, observed planning input; **no synthetic incidence is used.**
- **Designs.** All eight grid-study designs plus Simple Random Sampling (SRS) were adapted to the irregular 58-cluster map (graph checkerboard, regional saturation, spatial blocks, etc.).
- **Model.** Simulated education outcomes follow the same spatial-lag model as the grid study with a known effect ($\tau = 1$), spatial dependence $\rho \in \{0, 0.2, 0.5\}$, spillover $\gamma \in \{0.5, 0.8\}$, and two spillover regimes (control-only, both arms). **Incidence directs allocation only; it does not enter the education outcome** ($\beta = 0$).
- **Scale and quality.** 1,248 reported settings from 936 distinct simulations and 7,598,000 independent fits, plus 2,280,000 tail-refinement outcomes. Every setting is complete and passes prespecified precision targets for simulation error, with no warnings, aliasing, boundary fits or failures.

Key results (queen contiguity, primary model):

1. **Control-only spillover — saturation wins.** Saturation Quadrants lowered mean squared error (MSE) by 27.4% relative to SRS; incidence-guided saturation by 21.1%–29.9% across years. Both beat SRS in every parameter setting, and at the settings examined for downside ($\rho = 0, 0.5$; $\gamma = 0.8$) both also reduced the *downside* (90th percentile and worst-10% mean of allocation MSE) by about 21–43%.
2. **Both-arms spillover — SRS is competitive.** SRS (MSE 0.0764), balancing and saturation designs performed about equally; no design is clearly better.
3. **Designs to avoid.** Checkerboard was never better than SRS; blocking was never clearly better and was up to 1.4× worse under control-only spillover; Isolation Buffer treats far fewer clusters and is poor under both-arms spillover; High Incidence Focus is a single deterministic allocation whose accuracy depends on the outcome model.
4. **Coverage** of nominal 95% intervals is 92.6%–93.6% in every design, slightly below nominal (as in the grid study).

Bottom line. The application reproduces the grid study's central, regime-dependent conclusion on real geography and real incidence: *saturation designs are the recommended choice when spillover is expected mainly in control clusters, and SRS remains a reasonable default when spillover reaches both arms.* Incidence-guided saturation was about as accurate as Saturation Quadrants, so incidence can steer the highest treatment saturations to the highest-burden regions at little or no cost in precision.

How to read this report. Sections 1–8 (the main body) tell the full story. Section 9 holds additional results for reference. The final section lists key decisions made.

1 Background: what the grid simulation showed

The grid study compared treatment assignment designs for a spatial cluster randomized trial (CRT) on a 10×10 grid of 100 clusters. Outcomes followed a spatial-lag model,

$$Y = (I - \rho W)^{-1} [\tau Z + S(Z) + \beta X + \epsilon], \qquad \epsilon \sim N(0, \sigma^2 I),$$

where Z is treatment, X is baseline incidence (three generating mechanisms), W is a row-standardized contiguity matrix (queen primary, rook sensitivity), and $S(Z)$ is spillover of magnitude γ reaching either **control clusters only** or **both arms**. The eight designs were evaluated over 12,800 scenarios (1,600 per design, plus 1,600 for an SRS benchmark), fitting the spatial-lag model by maximum likelihood with the true spillover term.

Table 1: Grid study benchmark at $\tau = 1$, queen contiguity: mean MSE over 80 scenarios per regime (five incidence configurations $\times$ four ρ $\times$ four γ).

	Both arms		Control only	
Design	MSE	Ratio to SRS	MSE	Ratio to SRS
Simple Random Sampling	0.0424	1.00	0.2146	1.00
Incidence-Guided Saturation	0.0459	1.08	0.1355	0.63
Saturation Quadrants	0.0442	1.04	0.1359	0.63
Balanced Quartiles	0.0421	0.99	0.2195	1.02
Balanced Halves	0.0420	0.99	0.2179	1.02
2×2 Blocking	0.0623	1.47	0.5761	2.68
Checkerboard	0.0624	1.47	2.1114	9.84
Isolation Buffer	0.1608	3.80	0.1595	0.74
High Incidence Focus	0.1617	3.82	0.3292	1.53

The grid study's findings that motivate the application:

- **Spillover regime decides the ranking.** Under control-only spillover, Saturation Quadrants and Incidence-Guided Saturation had the lowest MSE (about 0.63 times SRS); checkerboard (9.8×) and 2×2 blocking (2.7×) failed badly, plausibly because fine interspersion leaves little information to separate treatment from spillover. Under both-arms spillover, SRS, Balanced Quartiles/Halves and saturation were within about 8% of one another.
- **Average MSE is not the whole story.** A trial realizes one allocation. A focused pilot showed that SRS's worst 10% of allocations were much worse than its mean under control-only spillover, while saturation reduced that downside.
- **Open question for the application:** do these conclusions survive an irregular real map, unequal cluster populations and *observed* incidence?

2 Data and preparation

Sources. County-year SUD death counts come from North Carolina death-certificate data prepared by A. Habib (restricted; not publicly available). The numerator uses Habib's corrected case filtering, which retains heart-failure deaths that an earlier filter excluded as presumed non-sudden; the earlier counts (21,147 deaths) are kept only to reconcile with Habib's unpublished report. The denominator is the SEER population aged 18–64 for each county and year.

Rates. For cluster i and year y, annual incidence is $100{,}000\, D_{iy}/P_{iy}$; pooled incidence is $100{,}000 \sum_y D_{iy} / \sum_y P_{iy}$ (per 100,000 person-years). All 100 counties and 58 clusters are present in every year, and counts and populations sum back to statewide totals.

Rates rose over the period, especially in 2020–2021, and cluster rankings are fairly but not perfectly stable from year to year. The analysis therefore treats **each year as a separate planning scenario** rather than averaging years, so that the incidence-guided designs are evaluated on four realistic but different inputs.

Table 2: Observed SUD incidence per 100,000 adults aged 18–64: statewide and across the 58 service-area clusters. Only rates are shown; restricted counts are not reported.

Period	Statewide rate	Cluster min	Cluster median	Cluster max
2018	83.9	42.6	108.3	199.4
2019	85.9	43.0	111.4	169.3
2020	97.1	47.2	127.4	268.0
2021	100.5	48.4	132.3	234.0
2018–2021 pooled	91.9	45.3	125.6	194.9

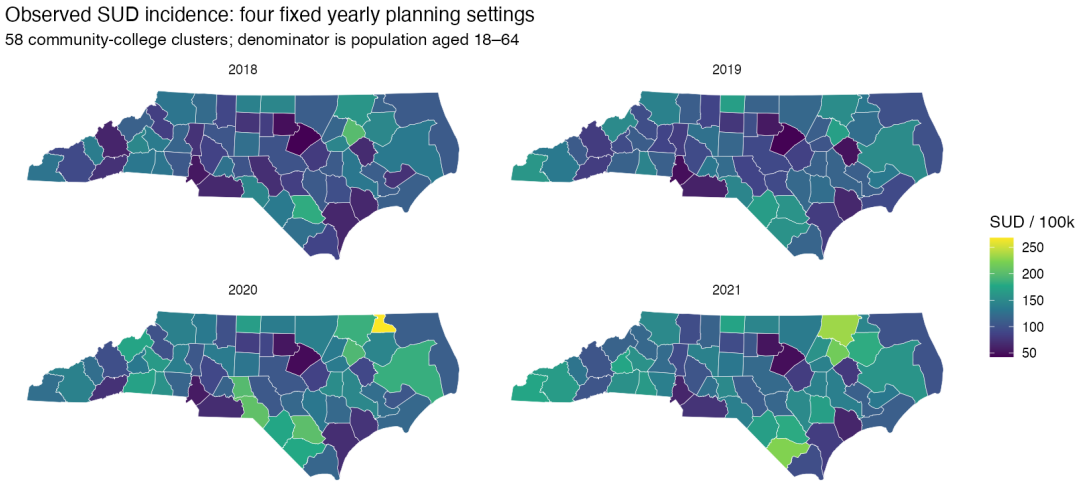


Figure 1: Observed annual SUD incidence per 100,000 adults aged 18–64 in the 58 service-area clusters. These fixed inputs direct incidence-aware allocation; they are not simulated outcomes or measures of education ability.

3 Clusters and spatial weights

Clusters. The randomization units are 58 community-college service areas that partition North Carolina's 100 counties. Three counties served by more than one college were assigned by convention: Northampton to Halifax Community College, Bertie to Martin Community College, and Hertford alone to Roanoke-Chowan Community College. These are analysis conventions, not exclusive service commitments.

Spatial weights. Adjacency comes from legal county boundaries dissolved to service areas. Queen contiguity (clusters sharing any boundary point are neighbors) is primary, and rook contiguity (sharing a boundary segment) is a sensitivity analysis. Queen degree ranges from 1 to 8 (mean 4.93). Legal (not shoreline-clipped) boundaries were chosen deliberately: clusters facing each other across sounds and rivers count as neighbors, because an education intervention can plausibly reach across water. Adjacency is a proxy for professional communication, not a measured network.

4 Adapting the designs to the NC map

The grid designs rely on rows, columns and quadrants that do not exist on an irregular map. Each was adapted to keep its *allocation mechanism* while respecting the geography. All partitions were fixed **before** looking at any simulated performance.

Three adaptations matter for interpretation:

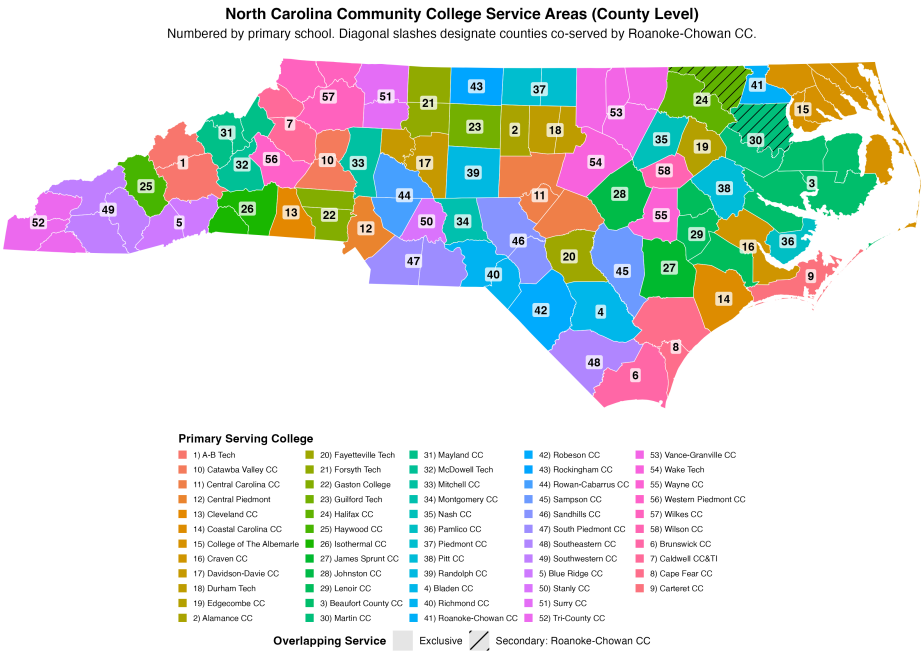


Figure 2: The 58 community-college service-area clusters (counties colored by primary serving college). Hatching marks Northampton and Bertie counties, which also lie in Roanoke-Chowan Community College's wider network.

Table 3: How each design was adapted from the grid to the 58-cluster NC map.

Design	Grid rule	NC adaptation
Simple Random Sampling	Random 50 of 100	Random 29 of 58
Saturation Quadrants	Random 20/40/60/80% per quadrant	Same saturations on 4 fixed geographic regions (k-means)
Incidence-Guided Saturation	80/60/40/20% by quadrant mean incidence	Regions ranked by mean within-region incidence rank
Balanced Quartiles	Equal treatment within incidence quartiles	Same; 29 treated, random extra in odd strata
Balanced Halves	Equal treatment within incidence halves	14/15 split; random choice of the 15-treated half
2×2 Blocking	2 of 4 in each 2x2 block	Half of each of 14 irregular k-means groups (one singleton stays control)
Checkerboard	Alternating (x+y) mod 2	Deterministic greedy graph coloring by degree
Isolation Buffer	Random maximal independent set	Same, on the NC queen graph
High Incidence Focus	Treat 50 highest-incidence	Treat 29 highest-rate clusters (deterministic)

- **Two designs become deterministic.** Graph checkerboard and High Incidence Focus each produce one allocation per year. Their accuracy is conditional on the assumed outcome model, not protected by randomization.
- **Regions and blocks are irregular.** Saturation uses one fixed four-region partition for all years, chosen from 250 geographic k-means candidates by compactness and balance in population, cluster and county counts. Spatial blocking uses 14 groups per year; one singleton group in each year is always control (rounding 0.5 down).
- **Budgets differ.** Most designs treat 29 clusters (about half of the population), but treated *population* varies, so the study reports both (Table 4).

5 Methods

Outcome model. For the education pilot, the primary model sets $\beta = 0$:

$$Y = (I - \rho W)^{-1} [\tau Z + S(Z) + \epsilon], \qquad \tau = 1,\ \sigma = 1,$$

Table 4: Treatment budgets: mean treated clusters / percent of the year's population aged 18–64. Population share measures geographic reach, not the number of professionals educated.

Design	2018	2019	2020	2021
Simple Random Sampling	29.0 / 49.5%	29.0 / 49.5%	29.0 / 49.5%	29.0 / 49.5%
Incidence-Guided Saturation	29.0 / 53.2%	29.0 / 45.1%	29.0 / 47.9%	29.0 / 51.8%
Saturation Quadrants	29.0 / 49.2%	29.0 / 49.2%	29.0 / 49.1%	29.0 / 49.1%
Balanced Quartiles	29.0 / 48.5%	29.0 / 49.7%	29.0 / 50.6%	29.0 / 49.8%
Balanced Halves	29.0 / 50.0%	29.0 / 50.5%	29.0 / 49.2%	29.0 / 49.4%
2×2 Blocking	28.0 / 50.0%	31.0 / 55.1%	27.0 / 48.0%	29.0 / 50.3%
Checkerboard	27.0 / 50.4%	27.0 / 50.4%	27.0 / 50.4%	27.0 / 50.7%
Isolation Buffer	15.9 / 26.7%	15.9 / 26.7%	15.9 / 26.7%	15.9 / 26.7%
High Incidence Focus	29.0 / 22.2%	29.0 / 23.5%	29.0 / 23.3%	29.0 / 22.4%

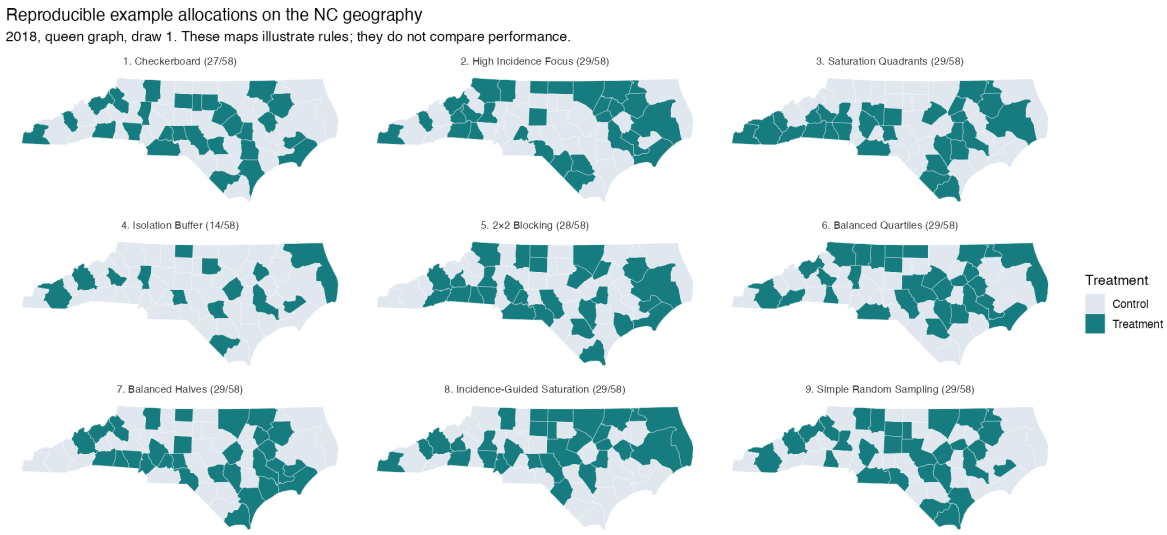


Figure 3: Example allocations under each design (2018 inputs). Treated clusters are shaded. On the NC map, Saturation Quadrants and Incidence-Guided Saturation use four fixed regions and 2×2 Blocking uses spatial blocks.

with the 58-cluster W. Observed incidence affects *who is treated* (for incidence-aware designs) but neither baseline education nor the treatment effect. This reflects the substantive premise that SUD incidence is a reason to target the intervention, not a predictor of education ability. A matched hypothetical sensitivity with $\beta = 1$ (rank-scaled incidence in both the outcome and the fit) is reported in Section 9.

Estimation. τ is estimated by maximum likelihood for the spatial-lag model with an intercept, Z and the true regime-specific spillover term, using the validated estimation engine from the grid study (cross-checked against `spatialreg::lagsarlm` on the NC weights).

Parameter grid. Queen weights; $\rho \in \{0, 0.2, 0.5\}$; $\gamma \in \{0.5, 0.8\}$; two spillover regimes; four years; nine designs, giving 432 primary settings. Sensitivity analyses (rook weights at $\rho \in \{0, 0.5\}$ and $\gamma \in \{0.5, 0.8\}$; $\beta = 1$; alternative regional summaries) bring the total to 1,248 reported settings.

Monte Carlo design. For each setting, $J = 100$ allocations are drawn and $R = 100$ independent outcomes simulated per unique allocation (deterministic designs: $R = 1{,}000$ outcomes for their single allocation). Because each result is estimated from a finite number of simulated datasets, it carries simulation (Monte Carlo) error; its Monte Carlo standard error (MCSE) measures this uncertainty and shrinks as more datasets are simulated. Replication was extended until the MCSE of mean MSE was at most 5% of its value and the MCSE of coverage at most 0.01; every setting passed. Every random draw used a fixed, reproducible seed.

Performance measures. Mean MSE, bias and 95% confidence interval coverage for $\hat{\tau}$, each compared with SRS in the same setting. Because a real trial realizes *one* allocation, we also estimate each allocation's risk $m(a) = E_\epsilon[(\hat{\tau} - \tau)^2 \mid a]$ and summarize its distribution across allocations by the 90th percentile (q90) and the mean of the worst 10% of allocations. A separate tail refinement re-used the same allocations with $R = 400$ outcomes at $\rho \in \{0, 0.5\}$, $\gamma = 0.8$.

6 Results

6.1 Average accuracy

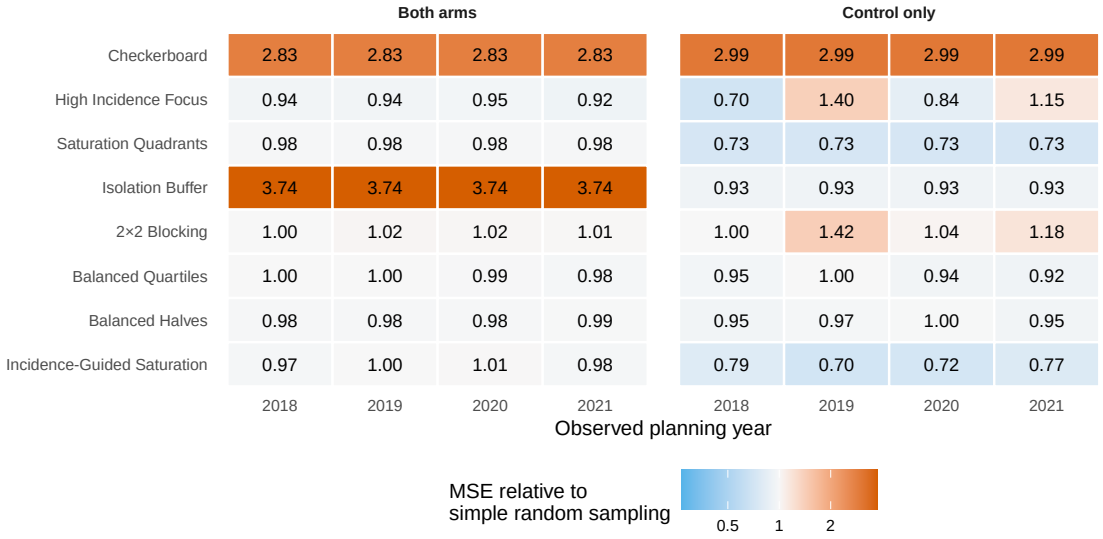


Figure 4: Yearly mean MSE relative to SRS in the same year and regime (queen; six ρ/γ settings averaged equally). Blue = lower error than SRS. SRS mean MSE is 0.07638 (both arms) and 0.30761 (control only).

Table 5: Primary results by design and spillover regime. MSE and ratio give the range over the four planning years; coverage is the four-year mean. Designs whose allocation does not use incidence (SRS, Saturation Quadrants, checkerboard, buffer) have identical results across years.

	Control-only spillover			Both-arms spillover		
Design	MSE	Ratio to SRS	Coverage	MSE	Ratio to SRS	Coverage
Simple Random Sampling	0.3076	1.00	0.929	0.0764	1.00	0.932
Incidence-Guided Saturation	0.2156–0.2429	0.70–0.79	0.927	0.0743–0.0768	0.97–1.01	0.935
Saturation Quadrants	0.2233	0.73	0.926	0.0752	0.98	0.935
Balanced Quartiles	0.2844–0.3062	0.92–1.00	0.929	0.0745–0.0762	0.98–1.00	0.933
Balanced Halves	0.2909–0.3061	0.95–1.00	0.930	0.0746–0.0758	0.98–0.99	0.933
2×2 Blocking	0.3062–0.4363	1.00–1.42	0.929	0.0761–0.0781	1.00–1.02	0.932
Checkerboard	0.9204	2.99	0.933	0.2159	2.83	0.936
Isolation Buffer	0.2857	0.93	0.931	0.2859	3.74	0.932
High Incidence Focus	0.2143–0.4302	0.70–1.40	0.931	0.0703–0.0729	0.92–0.95	0.934

- **Control-only spillover.** Saturation Quadrants (0.2233 vs SRS 0.3076) and incidence-guided saturation were the most accurate designs averaged over years (High Incidence Focus was better in 2018 alone, ratio 0.70, but ranged up to 1.40). Both had lower MSE than SRS in all 24 year-by-setting combinations, with the upper MCSE-based bound of every difference below zero (48 of 48 comparisons). The two saturation designs performed similarly (incidence-guided yearly MSE within about −3% to +9% of Saturation Quadrants; equal on the grid), so **incidence guidance targets high-burden regions at little or no cost in accuracy.** Balanced designs and Isolation Buffer were close to or slightly below SRS; checkerboard (about 3× SRS) was much worse, and blocking and High Incidence Focus were worse in some years.

- **Both-arms spillover.** SRS, the balanced designs, blocking and both saturation designs were within a few percent of each other; small differences should not be read as equivalence or superiority. Checkerboard (about 2.8×) and Isolation Buffer (about 3.7×) were far worse. High Incidence Focus had slightly lower MSE than SRS, but it is one deterministic allocation (see Section 7).
- **Bias and coverage.** Bias was small throughout ($|\text{bias}| \leq 0.05$). Coverage averaged about 93.1%, slightly below 95% for every design, as in the grid study. Passing the precision gates does not repair this finite-sample undercoverage.

6.2 Downside risk across allocations

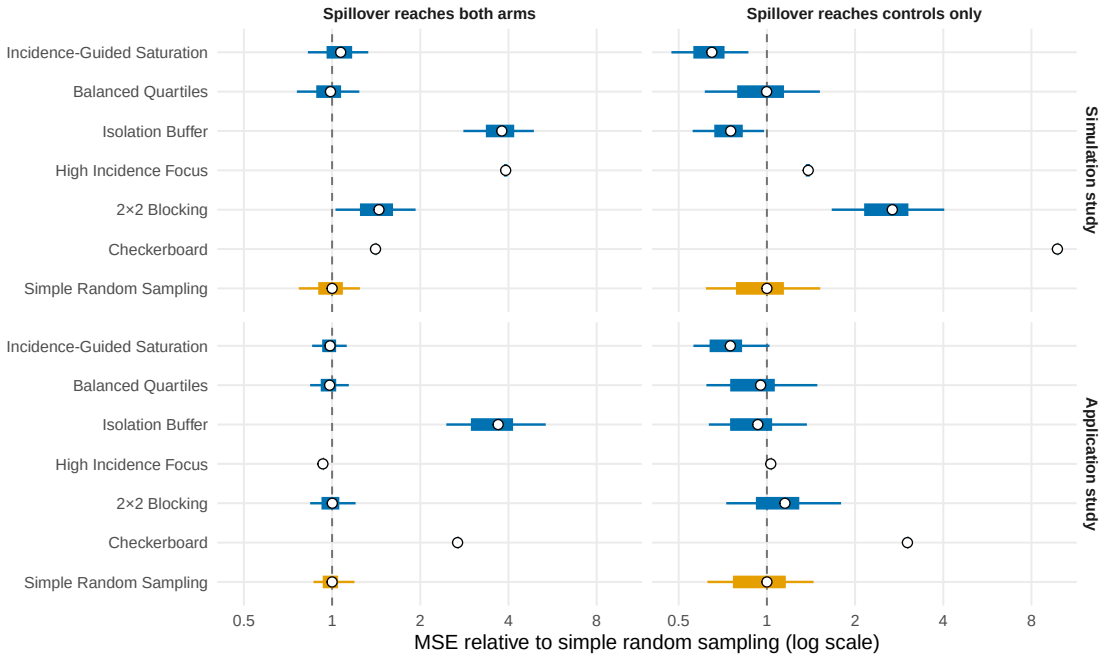


Figure 5: Distribution of allocation-specific MSE relative to SRS mean MSE, for six designs and SRS. Top: focused grid pilot (simulation study); bottom: refined NC settings ($\rho = 0, 0.5$; $\gamma = 0.8$; all years). Columns are the spillover regimes; SRS is in orange. Thin lines: 5th–95th percentiles; thick bars: 25th–75th; circles: mean. Deterministic designs appear as a single point.

Under **control-only** spillover, Saturation Quadrants's estimated q90 was 31.8%–32.9% below SRS and its worst-decile mean 34.6%–37.8% below; incidence-guided saturation's reductions were 20.8%–38.6% and 20.7%–43.4%. The pattern held when one half of the outcomes picked the worst allocations and the other half evaluated them. So saturation lowers both the average error and the chance of drawing a poor allocation. Under **both-arms** spillover, allocation-to-allocation spread was small for SRS and the balancing and saturation designs (Isolation Buffer's was wide), and tail differences among them were within simulation noise.

These tail estimates are descriptive. With $J = 100$ allocations, the worst decile contains only ten draws; extra outcomes reduce noise in each allocation's risk but do not enlarge the allocation sample.

6.3 Reach

High Incidence Focus treats 29 clusters but only about 22–24% of the working-age population (its high-rate clusters tend to have smaller populations); Isolation Buffer treats about 16 clusters and 27% of the population. All other designs reach roughly half the population (Table 4).

7 Comparison with the grid findings

Table 6: Mean MSE relative to SRS: grid benchmark ($\tau = 1$, queen) versus NC application (mean over years). Values below 1 mean lower error than SRS.

	Control-only		Both arms	
Design	Grid	NC	Grid	NC
Incidence-Guided Saturation	0.63	0.74	1.08	0.99
Saturation Quadrants	0.63	0.73	1.04	0.98
Balanced Quartiles	1.02	0.95	0.99	0.99
2×2 Blocking	2.68	1.16	1.47	1.01
Checkerboard	9.84	2.99	1.47	2.83
Isolation Buffer	0.74	0.93	3.80	3.74
High Incidence Focus	1.53	1.02	3.82	0.94

What held.

- **Regime dependence.** In both studies the spillover regime, not the incidence pattern, determined which designs did well.
- **Saturation under control-only spillover.** Saturation Quadrants and Incidence-Guided Saturation were the best designs under control-only spillover in both studies, on average and in the downside.
- **Balancing tracks SRS.** Balanced Quartiles/Halves were indistinguishable from SRS in both.
- **Fine interspersion fails under control-only spillover.** Checkerboard did poorly in both studies; blocking was worse than SRS on the grid and in some NC years.

What changed, and why.

- **Saturation's advantage is smaller in NC** (about 0.70–0.79× SRS vs about 0.63 on the grid). A plausible explanation, not tested directly, is that four irregular regions of unequal size give less contrast in local treatment intensity than the grid's equal quadrants.
- **Blocking is much closer to SRS in NC** (about 1.0–1.4× under control-only spillover vs 2.7× on the grid). Plausibly, the 14 irregular spatial groups are larger and looser than 2×2 squares, so treated and control clusters are less finely interspersed.
- **Checkerboard changed in both directions.** Under control-only spillover it is less extreme (about 3× SRS vs about 10×), likely because a graph coloring cannot alternate perfectly on an irregular map; under both arms it is worse (about 2.8× vs 1.5×). Its effect is identified on the NC graph but estimated imprecisely.
- **High Incidence Focus looks much better in NC under both arms** (about SRS level vs 3.8× on the grid). Under $\beta = 0$ the incidence it targets is unrelated to the outcome, so a fixed, roughly balanced-size allocation does fine. When outcomes are related to incidence ($\beta = 1$ sensitivity), its MSE rises 1.4–4.4-fold across settings. Its NC result is therefore model-dependent, not robust.
- **Isolation Buffer** remains poor under both-arms spillover (about 3.7–3.8× SRS in both studies) and treats only about 16 clusters.

8 Implications and limitations

For the education pilot.

- **Control-only spillover plausible:** if spillover is expected mainly among untreated clusters (e.g., professionals in control areas hearing about the training), a saturation design (Saturation Quadrants or Incidence-Guided Saturation) is the recommended choice. It lowers average error by about a quarter and reduces the chance of a poor allocation.
- **Both-arms spillover expected:** SRS is a reasonable, simple default, with balanced and saturation designs as equivalent alternatives.
- **Incidence guidance:** incidence-guided saturation treats high-burden regions most intensively while keeping about the same accuracy as Saturation Quadrants, which makes it a natural choice when reaching high-burden areas is also a program goal.

For the method. The application shows that the grid study's design guidance transfers to a real, irregular geography with unequal populations and observed incidence, once designs are adapted with prespecified rules. The key design question for a trial team is the expected spillover regime, not the incidence pattern.

Limitations.

- **Simulation, not efficacy.** These are design-stage simulations with a known spillover form and constant effect. They do not show that education reduces SUD.
- **Population is not the sample.** Working-age population is a geographic denominator, not the number of healthcare professionals educated.
- **Fixed incidence inputs.** Historical rates are held fixed; measurement error and change between planning and implementation are not propagated.
- **Undercoverage.** Coverage of about 93% is below nominal and would need calibration in an actual protocol.
- **Tail uncertainty.** Allocation-tail estimates rest on 100 sampled allocations per setting.
- **Unequal budgets.** Budgets differ by design; this is an evaluation of retained strategies, not an equal-cost optimization.

9 Additional results

9.1 Setting-level ratios to SRS

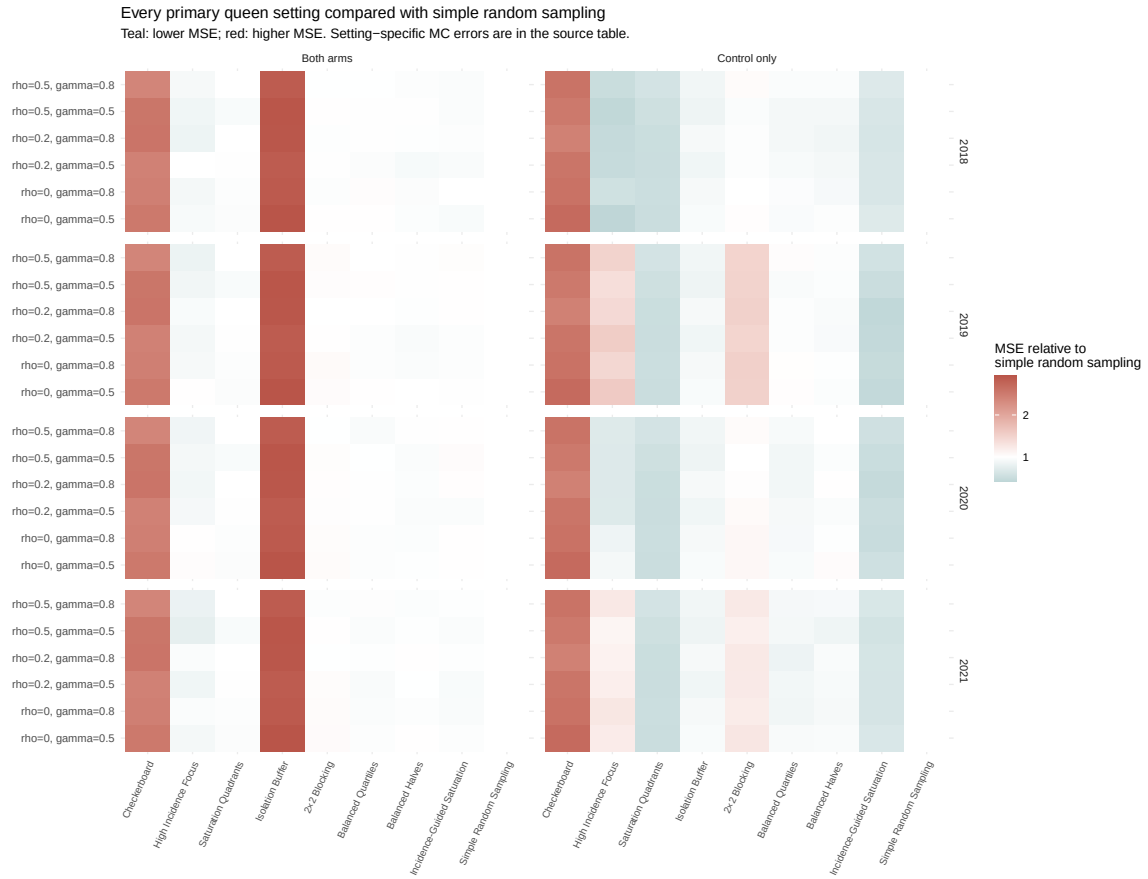


Figure 6: Setting-specific mean MSE ratios to year-matched SRS for each ρ/γ setting, year and regime (queen).

9.2 Sensitivity analyses

- $\beta = 1$ **(incidence in the outcome, adjusted in the fit).** High Incidence Focus is most affected: its treatment is aligned with the adjusted covariate, leaving little residual information about τ. Other designs change by a few

Table 7: Sensitivity MSE as a ratio to the primary-model MSE at exactly matched year, ρ, γ and regime (range over years). The rook comparison covers $\rho \in \{0, 0.5\}$ and $\gamma \in \{0.5, 0.8\}$.

	$\beta = 1$, incidence-adjusted		Rook weights	
Design	Control only	Both arms	Control only	Both arms
Simple Random Sampling	1.01–1.02	1.00–1.02	1.01	0.99
Incidence-Guided Saturation	1.02–1.04	1.01–1.04	0.97–1.04	0.98–1.00
Saturation Quadrants	1.01–1.02	1.02–1.02	1.00	1.01
Balanced Quartiles	1.02–1.03	1.00–1.00	0.94–1.04	0.98–1.01
Balanced Halves	1.02–1.03	1.00–1.02	0.91–1.01	0.99–1.00
2×2 Blocking	0.98–1.01	0.99–1.02	0.92–1.01	0.99–1.01
Checkerboard	0.99–1.05	1.03–1.22	1.23	1.15
Isolation Buffer	1.02–1.04	1.02–1.04	1.02	1.03
High Incidence Focus	1.55–2.36	3.91–4.14	0.89–1.04	1.01–1.07

percent (checkerboard by up to about 22% under both arms). This changes the baseline and the adjustment together; it is not a claim that SUD incidence predicts education ability.

- **Rook weights.** At matched corners, rook mainly raises checkerboard MSE (about 15–23% on yearly averages); other designs change by up to about 11%, and the regime-specific conclusions are unchanged.
- **Regional summaries (incidence-guided saturation only).** Ranking regions by mean raw rate or pooled deaths/population sometimes changes their order; matched MSE ratios range 0.91–1.13. The prespecified mean-rank rule is retained rather than chosen by performance.

9.3 Tail-refinement detail

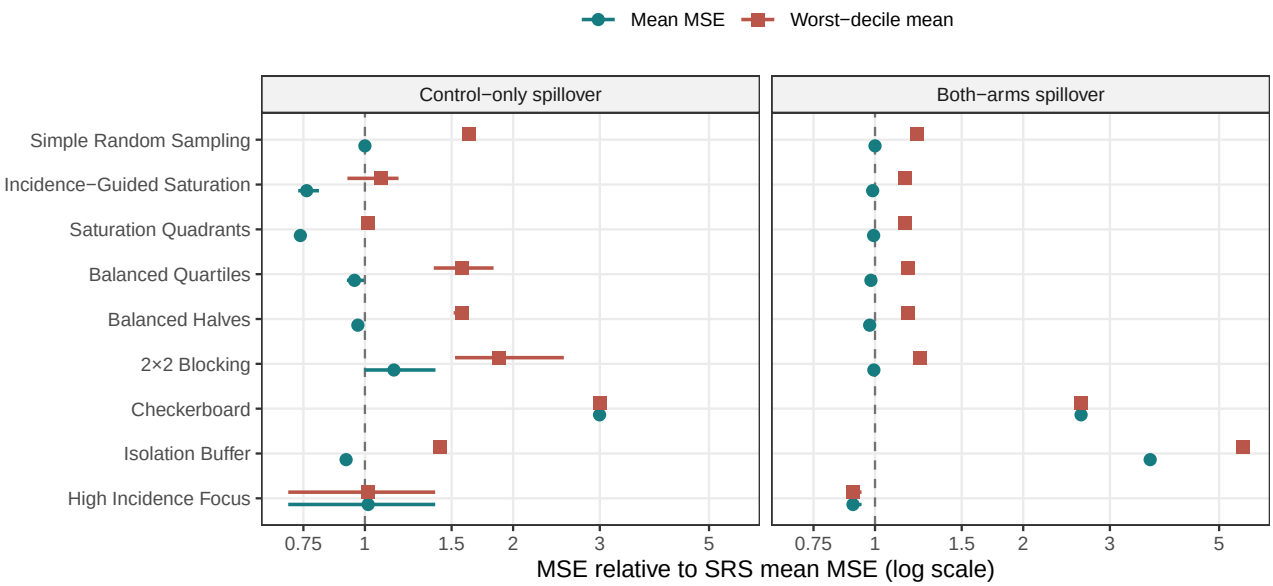


Figure 7: Average error and downside at queen $\rho = 0.5$, $\gamma = 0.8$, relative to the same year's SRS mean MSE (dashed line; log scale). Circles: mean MSE; squares: mean of the worst 10% of allocations (estimated risk, $R \geq 400$ outcomes per allocation; 1,000 for single-allocation designs). Points are four-year means; bars span the four planning years. Checkerboard and High Incidence Focus have one allocation, so mean and worst decile coincide.

All 112 refined settings with more than one possible assignment flag uncertain membership at the worst-decile boundary. For the two saturation designs under both-arms spillover, the worst-decile allocations chosen by the two outcome halves overlap little (Jaccard index, a 0–1 overlap measure, of 0–0.25), and about 61–86% of the apparent allocation variation is outcome noise. Small both-arms tail differences are therefore not evidence of superiority.

9.4 Population reach

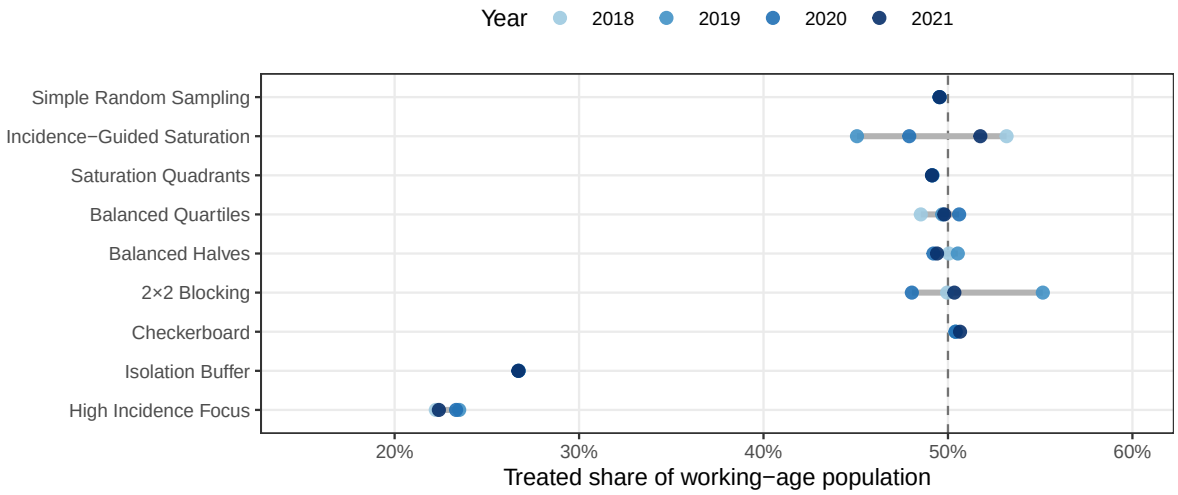


Figure 8: Share of the working-age population (ages 18–64) living in treated clusters, by design and planning year (mean over sampled allocations). The dashed line marks half of the population. Bars span the four years.

Key decisions made

1. $\beta = 0$ **primary.** Incidence directs allocation but is not modeled as affecting education outcomes; $\beta = 1$ is a labeled hypothetical sensitivity.
2. **Each year analyzed separately** rather than pooled, so incidence-guided designs face four realistic inputs.
3. **Geographic adaptations fixed in advance**: one four-region saturation partition; 14 yearly spatial blocks (one singleton always control); Balanced Halves 14/15 split; deterministic graph checkerboard.
4. **Unequal budgets disclosed, not forced equal.** Reach is reported alongside accuracy.
5. **High Incidence Focus** is deterministic for fixed inputs; reported but not recommended because its accuracy depends on the outcome model.
6. **Coverage reported as observed** (about 93%) rather than switching estimators after seeing results.
7. **Recommendations are regime-specific** rather than naming one overall winner.
8. **CTJ presentation:** main text shows six designs + SRS; Saturation Quadrants and Balanced Halves are in the supplement (they performed like incidence-guided saturation and Balanced Quartiles).